

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
```

```
x = np.array([2.1,2.3,2.5,3.0,3.2,3.9,4.0,4.2,4.7,5.0,3.9])
y = np.array([49000,56000,47700,np.nan,49800,
              67700,70300,65500,68200,73400,67700])
df= pd.DataFrame({
    "Experience":x,
    "Salary":y})
print(df)
```

	Experience	Salary
0	2.1	49000.0
1	2.3	56000.0
2	2.5	47700.0
3	3.0	NaN
4	3.2	49800.0
5	3.9	67700.0
6	4.0	70300.0
7	4.2	65500.0
8	4.7	68200.0
9	5.0	73400.0
10	3.9	67700.0

```
plt.plot(x,y,marker = 'D')
plt.xlabel("Experience")
plt.ylabel("Salary")
plt.title("Original Data - Experience Vs Salary")
plt.grid()
plt.show()
```



```
df.mean()
```

	0
Experience	3.49
Salary	60913.00

```
dtype: float64
```

```
df.median()
```

```
      0
Experience    3.55
Salary      63515.00
dtype: float64
```

```
df.describe()
```

	Experience	Salary
count	10.000000	10.000000
mean	3.490000	60913.000000
std	1.018114	9596.094171
min	2.100000	47700.000000
25%	2.625000	51350.000000
50%	3.550000	63515.000000
75%	4.150000	68075.000000
max	5.000000	73400.000000

```
df.isnull()
```

	Experience	Salary
0	False	False
1	False	False
2	False	False
3	False	True
4	False	False
5	False	False
6	False	False
7	False	False
8	False	False
9	False	False
10	False	False

```
df.duplicated()
```

```
      0
0  False
1  False
2  False
3  False
4  False
5  False
6  False
7  False
8  False
9  False
10  True
dtype: bool
```

```
print("\n Missing Values")
print(df.isnull().sum())
```

```
Missing Values
Experience      0
```

```
Salary      1
dtype: int64
```

```
print("\n Duplicate Values")
print(df.duplicated().sum())
```

```
Duplicate Values
1
```

```
df['Salary'] = df["Salary"].fillna(df["Salary"].mean())
df
```

	Experience	Salary
0	2.1	49000.0
1	2.3	56000.0
2	2.5	47700.0
3	3.0	61530.0
4	3.2	49800.0
5	3.9	67700.0
6	4.0	70300.0
7	4.2	65500.0
8	4.7	68200.0
9	5.0	73400.0
10	3.9	67700.0

Next steps: [New interactive sheet](#)

```
df = df.drop_duplicates()
df
```

	Experience	Salary
0	2.1	49000.0
1	2.3	56000.0
2	2.5	47700.0
3	3.0	61530.0
4	3.2	49800.0
5	3.9	67700.0
6	4.0	70300.0
7	4.2	65500.0
8	4.7	68200.0
9	5.0	73400.0

Next steps: [New interactive sheet](#)

```
print(df.isnull().sum())
print(df.duplicated().sum())
```

```
Experience    0
Salary        0
dtype: int64
0
```

```
plt.plot(x,y,marker = 'D')
plt.xlabel("Experience")
plt.ylabel("Salary")
plt.title("Original Data - Experience Vs Salary")
plt.grid()
plt.show()
```



```
x= df[["Experience"]]
y= df["Salary"]
x_train,x_test,y_train,y_test = train_test_split(
    x,y,test_size=0.2, random_state=1)
```

```
x_test.shape
```

```
(2, 1)
```

```
x_train.shape
```

```
(8, 1)
```

```
y_test.shape
```

```
(2,)
```

```
y_train.shape
```

```
(8,)
```

```
model = LinearRegression()
```

```
model.fit(x_train, y_train)
```

```
LinearRegression
LinearRegression()
```

```
y_pred = model.predict(x_test)
print("Prediction value",y_pred)
```

```
Prediction value [54103.84009942 72752.24523612]
```

```
r2 = r2_score(y_test, y_pred)
print("R2 Score:",r2)
```

```
R2 Score: 0.8745514564851364
```

```
experience = np.array([[3]])
salary = model.predict(experience)
print("Predicted Salary for 5 Years Experience:",salary[0])
```

```
Predicted Salary for 5 Years Experience: 57833.52112676056
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names,
warnings.warn(
```

```
plt.plot(x,y,label="Actual Salary")
plt.plot(x, model.predict(x),label="Regression Line")
```

```
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Linear Regression - Experience vs Salary")

plt.legend()
plt.grid()
plt.show()
```



```
df.std()
```

	θ
Experience	1.018114
Salary	9596.094171

dtype: float64

```
df.corr()
```

	Experience	Salary
Experience	1.000000	0.874639
Salary	0.874639	1.000000